

CPRE 431

M03 HW

Assignments will be submitted in PDF format via Canvas.

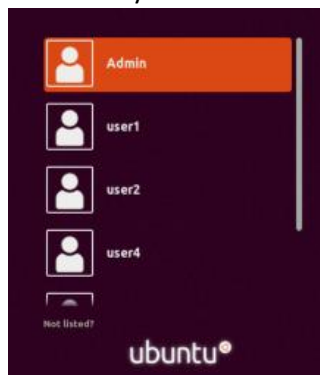
Please submit your homework online through Canvas. Late homework will not be accepted.

Please ensure that you support all your answers with the correct screenshots showing your solutions.

1. In this “lab” problem, you will be working on a Linux Server Virtual Machine (VM). An image of this VM is available on Canvas (attached with the HW). The VM is having an administrator and 5 users, as shown in the figure below. You don’t have access to any of the users of the Server, to be able to access the Server, you will need to perform password cracking! You were given a line of password hash from (/etc/shadow) for the administrator (admin1) of the Server (attached with the HW).
 - a. Determine the used hash type of the password.
\$6\$ represents SHA-512 hash was used for all users
 - b. Determine the salt value of the password.
xgL35S6 for Admin. It is the value between the second \$ and third \$.
 - c. Crack the passwords of all users using OpenSSL tool.
Admin: P@ssw0rd
user1: trustno1
user2: basketball
user3: pokemon
user4: batman
user5: qwerty12345

Hints:

- It would be useful if you search for Linux shadow file password format.
- You must use the latest OpenSSL version 1.1.1x for this problem. It would be helpful to read about creating Linux password hashes using OpenSSL.
- You can use a password list for your cracking. There is a password list of most used 100K passwords, according to NIST attached with the HW.
- You will need to write some code to iterate through the password list (feel free to use any language you prefer).
- After cracking the password, ensure that you can access the Server.



2. The given Server users are managed as follows:

- There are 5 users: user1, user2, ..., user5. The first four users are staff members, but user5 is an external consultant.
 - User1 and user2 are programmers only, user4 is a manager only, and user3 is both programmer and manager.
- a. You are now the administrator of the server, and you are responsible for managing users and groups. Create 3 groups named “allstaff”, “prog” (short for programmers) and “mgmt” (short for management). Add users to the corresponding groups. List the users and groups to ensure the correct previous setup of system users and groups. Explain with screenshots how you applied this.
- Inside their home directory, each programmer has a directory called code and a directory called documentation. Inside the code directory, there is a file called source_code.txt, as well as one application called myapp.exe. Inside the documentation directory, there is a file called notes.txt.
 - Each manager has a directory called finance (for financial information) including a confidential business.txt file.
 - Each user also has a file called schedule.txt in their home directory.

The below shows how I added allstaff, prog, and mgmt groups.

```
admin1@CPRE-431:~$ sudo groupadd allstaff
[sudo] password for admin1:
admin1@CPRE-431:~$ grep allstaff /etc/group
allstaff:x:1006:
admin1@CPRE-431:~$ tail /etc/group
gdm:x:125:
admin1:x:1000:
sambashare:x:126:admin1
vboxsf:x:999:
user1:x:1001:
user2:x:1002:
user3:x:1003:
user4:x:1004:
user5:x:1005:
allstaff:x:1006:

admin1@CPRE-431:~$ sudo groupadd prog
admin1@CPRE-431:~$ sudo groupadd mgmt
admin1@CPRE-431:~$ tail /etc/group
sambashare:x:126:admin1
vboxsf:x:999:
user1:x:1001:
user2:x:1002:
user3:x:1003:
user4:x:1004:
user5:x:1005:
allstaff:x:1006:
prog:x:1007:
mgmt:x:1008:
```

The below shows how I added the users to their respective groups.

```
admin1@CPRE-431:~$ sudo adduser user1 allstaff
Adding user 'user1' to group 'allstaff' ...
Adding user user1 to group allstaff
Done.
admin1@CPRE-431:~$ sudo adduser user2 allstaff
Adding user 'user2' to group 'allstaff' ...
Adding user user2 to group allstaff
Done.
admin1@CPRE-431:~$ sudo adduser user3 allstaff
Adding user 'user3' to group 'allstaff' ...
Adding user user3 to group allstaff
Done.
admin1@CPRE-431:~$ sudo adduser user4 allstaff
Adding user 'user4' to group 'allstaff' ...
Adding user user4 to group allstaff
Done.

admin1@CPRE-431:~$ sudo adduser user1 prog
Adding user 'user1' to group 'prog' ...
Adding user user1 to group prog
Done.
admin1@CPRE-431:~$ sudo adduser user2 prog
Adding user 'user2' to group 'prog' ...
Adding user user2 to group prog
Done.
admin1@CPRE-431:~$ sudo adduser user4 mgmt
Adding user 'user4' to group 'mgmt' ...
Adding user user4 to group mgmt
Done.
admin1@CPRE-431:~$ sudo adduser user3 mgmt
Adding user 'user3' to group 'mgmt' ...
Adding user user3 to group mgmt
Done.
admin1@CPRE-431:~$ sudo adduser user3 prog
Adding user 'user3' to group 'prog' ...
Adding user user3 to group prog
Done.
```

We are also able to confirm that the users were allocated to the correct groups by using the command (sudo cat /etc/group) which shows the below output at the end of the file:

```
allstaff:x:1006:user1,user2,user3,user4
prog:x:1007:user1,user2,user3
mgmt:x:1008:user4,user3
```

- b. Configure file access controls so that it explicitly applies only the following:
- All users can view each other's schedules, but not other files in their home directory (except for as stated in the following).
 - All staff can view files in the documentation directory of other staff.
 - Programmers can view and edit each other's source code files, create new files in any code directory, as well as run each other's myapp.exe files.
 - Financial information (in the finance directory) is only viewable by the manager that owns the files, not by any other user.

For step 1, I used (chmod u=r,go=r schedule.txt), all other users also able to view each others schedules and the upcoming steps ensure that other files in each users home directory is only accessible to the users that need the permission.

For step 2, I used (chgrp -R allstaff documentation/) and (chmod 750 documentation/) for the documentation directory for all users 1-4 and also added (chmod ug=r,o=- notes.txt) for the notes.txt file.

For step 3, I used (chgrp -R prog code/) and (chmod -R g+rw code/) to be able to read, write, execute within the code/ directory.

For step 4, I used (chmod -R go-rwx finance/) in user 3 and user 4 ensuring only they are able to view their files and also ensuring that only user 3 can view his own finance/ directory and user 4 can view his own finance/ directory.

The below are the checks conducted to ensure all the permissions are given correctly.

User 5 only able to see schedule.txt and nothing else since he is not a staff.

```
user5@CPRE-431:~$ cd ..
user5@CPRE-431:/home$ cd user1
user5@CPRE-431:/home/user1$ ls
code  documentation  Downloads  Pictures  schedule.txt  Videos
Desktop  Documents  Music  Public  Templates
user5@CPRE-431:/home/user1$ cat schedule.txt
user5@CPRE-431:/home/user1$ cd code/
bash: cd: code/: Permission denied
user5@CPRE-431:/home/user1$ cd documentation/
bash: cd: documentation/: Permission denied
user5@CPRE-431:/home/user1$ cd /home/user2
user5@CPRE-431:/home/user2$ cat schedule.txt
user5@CPRE-431:/home/user2$ cd code/
bash: cd: code/: Permission denied
user5@CPRE-431:/home/user2$ cd documentation/
bash: cd: documentation/: Permission denied
user5@CPRE-431:/home/user2$ cd /home/user3
user5@CPRE-431:/home/user3$ cat schedule.txt
user5@CPRE-431:/home/user3$ cd code/
bash: cd: code/: Permission denied
user5@CPRE-431:/home/user3$ cd documentation/
bash: cd: documentation/: Permission denied
user5@CPRE-431:/home/user3$ cd /home/user4
user5@CPRE-431:/home/user4$ cat schedule.txt
user5@CPRE-431:/home/user4$ ls -l
total 36
drwx----- 2 user4 user4 4096 Sep 30 2019 Desktop
drwx----- 2 user4 user4 4096 Sep 30 2019 Documents
drwx----- 2 user4 user4 4096 Sep 30 2019 Downloads
drwx----- 2 user4 user4 4096 Sep 30 2019 Finance
drwx----- 2 user4 user4 4096 Sep 30 2019 Music
drwx----- 2 user4 user4 4096 Sep 30 2019 Pictures
drwx----- 2 user4 user4 4096 Sep 30 2019 Public
-rw-r--r-- 1 user4 user4 0 Sep 30 2019 schedule.txt
drwx----- 2 user4 user4 4096 Sep 30 2019 Templates
drwx----- 2 user4 user4 4096 Sep 30 2019 Videos
user5@CPRE-431:/home/user4$ cd finance/
bash: cd: finance/: Permission denied
user5@CPRE-431:/home/user4$ cd /home/user3/
user5@CPRE-431:/home/user3$ cd finance/
bash: cd: finance/: Permission denied
user5@CPRE-431:/home/user3$
```

User 1 Access Rights (able to access all the users schedules, all code directories, create and modify new files within code directory, but not able to enter finance directory of user3 and user4):

```
user1@CPRE-431:~$ cd /home/user2
user1@CPRE-431:/home/user2$ cat schedule.txt
user1@CPRE-431:/home/user2$ cd documentation/
user1@CPRE-431:/home/user2/documentation$ cat notes.txt
user1@CPRE-431:/home/user2/documentation$ cd ..
user1@CPRE-431:/home/user2$ cd code/
user1@CPRE-431:/home/user2/code$ cat
myapp.exe          source_code.txt
user1@CPRE-431:/home/user2/code$ cat source_code.txt
user1@CPRE-431:/home/user2/code$ nano source_code.txt
user1@CPRE-431:/home/user2/code$ cd /home/user3
user1@CPRE-431:/home/user3$ cat schedule.txt
user1@CPRE-431:/home/user3$ cd documentation/
user1@CPRE-431:/home/user3/documentation$ cat notes.txt
user1@CPRE-431:/home/user3/documentation$ cd ..
user1@CPRE-431:/home/user3$ cd code/
user1@CPRE-431:/home/user3/code$ cat source_code.txt
user1@CPRE-431:/home/user3/code$ cd /home/user4
user1@CPRE-431:/home/user4$ cd finance/
bash: cd: finance/: Permission denied
user1@CPRE-431:/home/user4$ ls
Desktop  Documents  Downloads  finance  Music  Pictures  Public  schedule.txt  Templates  Videos
user1@CPRE-431:/home/user4$ cat schedule.txt
user1@CPRE-431:/home/user4$ cd /home/user3
user1@CPRE-431:/home/user3$ cd finance/
bash: cd: finance/: Permission denied
user1@CPRE-431:/home/user3$
```

User 2 Access Rights (able to access all the users schedules, all code directories, create and modify new files within code directory, but not able to enter finance directory of user3 and user4):

```
user2@CPRE-431:~$ cd /home/user1
user2@CPRE-431:/home/user1$ cat schedule.txt
user2@CPRE-431:/home/user1$ cd documentation/
user2@CPRE-431:/home/user1/documentation$ cat notes.txt
user2@CPRE-431:/home/user1/documentation$ cd ..
user2@CPRE-431:/home/user1$ cd code/
user2@CPRE-431:/home/user1/code$ cat source_code.txt
user2@CPRE-431:/home/user1/code$ nano source_code.txt
user2@CPRE-431:/home/user1/code$ cd /home/user3
user2@CPRE-431:/home/user3$ cat schedule.txt
user2@CPRE-431:/home/user3$ cd documentation/
user2@CPRE-431:/home/user3/documentation$ cat notes.txt
user2@CPRE-431:/home/user3/documentation$ cd ..
user2@CPRE-431:/home/user3$ cd code/
user2@CPRE-431:/home/user3/code$ cat source_code.txt
user2@CPRE-431:/home/user3/code$ nano source_code.txt
user2@CPRE-431:/home/user3/code$ cd /home/user4
user2@CPRE-431:/home/user4$ cat schedule.txt
user2@CPRE-431:/home/user4$ cd finance/
bash: cd: finance/: Permission denied
user2@CPRE-431:/home/user4$ cd /home/user3
user2@CPRE-431:/home/user3$ cd finance/
bash: cd: finance/: Permission denied
user2@CPRE-431:/home/user3$
```


User 3 Access Rights (Since user 3 is a programmer and manager, he is able to access code directories for all users and the finance directory for his own user.)

```
user3@CPRE-431:~$ cd /home/user1
user3@CPRE-431:/home/user1$ cat schedule.txt
user3@CPRE-431:/home/user1$ cd documentation/
user3@CPRE-431:/home/user1/documentation$ cat notes.txt
user3@CPRE-431:/home/user1/documentation$ cd ..
user3@CPRE-431:/home/user1$ cd code/
user3@CPRE-431:/home/user1/code$ cat source_code.txt
user3@CPRE-431:/home/user1/code$ nano source_code.txt
user3@CPRE-431:/home/user1/code$ cd /home/user2
user3@CPRE-431:/home/user2$ cat schedule.txt
user3@CPRE-431:/home/user2$ cd documentation/
user3@CPRE-431:/home/user2/documentation$ cat notes.txt
user3@CPRE-431:/home/user2/documentation$ cd ..
user3@CPRE-431:/home/user2$ cd code/
user3@CPRE-431:/home/user2/code$ cat source_code.txt
user3@CPRE-431:/home/user2/code$ nano source_code.txt
user3@CPRE-431:/home/user2/code$ cd /home/user4
user3@CPRE-431:/home/user4$ cat schedule.txt
user3@CPRE-431:/home/user4$ cd finance/
bash: cd: finance/: Permission denied
user3@CPRE-431:/home/user4$ cd
user3@CPRE-431:~$ cd finance/
user3@CPRE-431:~/finance$ cat business.txt
user3@CPRE-431:~/finance$
```

User 4 Access Rights (Since user 4 is only a manager, he is able to access code directories for all users and the finance directory for his own user.)

```
user4@CPRE-431:~$ cd /home/user1
user4@CPRE-431:/home/user1$ cat schedule.txt
user4@CPRE-431:/home/user1$ cd documentation/
user4@CPRE-431:/home/user1/documentation$ cat notes.txt
user4@CPRE-431:/home/user1/documentation$ cd ..
user4@CPRE-431:/home/user1$ cd code/
bash: cd: code/: Permission denied
user4@CPRE-431:/home/user1$ cd /home/user2
user4@CPRE-431:/home/user2$ cat schedule.txt
user4@CPRE-431:/home/user2$ cd documentation/
user4@CPRE-431:/home/user2/documentation$ cat notes.txt
user4@CPRE-431:/home/user2/documentation$ cd ..
user4@CPRE-431:/home/user2$ cd code/
bash: cd: code/: Permission denied
user4@CPRE-431:/home/user2$ cd /home/user3
user4@CPRE-431:/home/user3$ cat schedule.txt
user4@CPRE-431:/home/user3$ cd documentation/
user4@CPRE-431:/home/user3/documentation$ cat notes.txt
user4@CPRE-431:/home/user3/documentation$ cd ..
user4@CPRE-431:/home/user3$ cd code/
bash: cd: code/: Permission denied
user4@CPRE-431:/home/user3$ cd finance/
bash: cd: finance/: Permission denied
user4@CPRE-431:/home/user3$ cd
user4@CPRE-431:~$ cd finance/
user4@CPRE-431:~/finance$ cat business.txt
user4@CPRE-431:~/finance$
```

If I missed something that you would like me to include to further clarify, please let me know.

Hints:

- Take screenshots of setting the access and listing it using (ls -l). Also, take screenshots of testing that the access control works by logging in as each user and checking they can(not) access the specified files/directories.
- Use only the basic Linux permissions. Do NOT use advanced permissions such as setfacl or getfacl.
- Use the ["Introduction to Linux" Page](#) to help you in the needed commands for this Homework.